of synthetic dyes, concern for environment worldwide and stringent environmental laws lead to the revival of natural dyes.

2.2.1.1 Advantages of Natural Dyes

- Natural dyes are extracted from natural sources and hence they are eco-friendly
- Produces soft and soothing colours
- These dyes provide excellent protection from UV rays
- Natural dyes like turmeric have anti-microbial properties and hence protect the fabrics and wearers from microbial attack
- Some natural dyes possess mosquito repellent and flame resistant property
- They can be obtained from the natural sources which are abundant in a particular area. Hence supply of raw materials will be continuous and transport charges will be lower

2.2.1.2 Disadvantages of Natural Dyes

- Natural dyes are difficult to store
- Dye extraction is a time consuming process
- Reproducibility of the same colour shade is difficult
- Impurities in natural dyes fades away the colour produced
- Availability of these dyes depends on the seasons
- Natural dyeing process is difficult to standardize

Classification of Natural Dyes

Natural dyes are classified in to three types based on the source of origin namely vegetable dyes, animal dyes and mineral dyes.

2.2.2 Vegetable Dyes

The earliest dyes were of vegetable origin, discovered by accidentally staining garments with juices of fruits or plants. Vegetable dyes are obtained from different parts of plants such as leaves, flowers, fruits, pods, bark etc. These vegetable dyes can be applied directly or with different mordants.

- **Indigo:** Indigo (blue dye) is called as the king of all natural dyestuffs. It imparts blue colour. It is extracted from the leaves of the leguminous plant, *Indigofera tinctoria*. It is suitable for dyeing cotton and wool.
- **Indian Madder:** It produces shades of red on textile fabrics. It is used for dyeing cotton and woollen fabrics. It is extracted from roots of *Rubia tinctoria.*
- **Turmeric:** It produces shades of yellow on fabrics. It is suitable for dyeing cotton, silk and wool. The yellow dye is extracted from the ground root (rhizome) of turmeric plant (*Curcuma longa*).
- **Marigold:** It is extracted from lemon or orange coloured marigold (*Calendula officinalis*) flower. It is suitable for dyeing both silk and wool fibres.
- **Henna:** The dye is extracted from the dried leaves of Henna plant, *Lawsonia inermis*. It produces yellowish orange colour. It is suitable for dyeing wool and silk fibres.
- **Tea:** Leaves of tea plants (*Camellia sinensis*) or tea powder is used to extract dye. It produces different shades of brown.
- **Onion:** The dye is extracted from the outer most skin or peel of the onion (*Allium cepa*). The onion skins if properly dried can be used for one year.

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